

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (EE)

December 2013

EE 708 Advanced Power Electronics in Power System

Date: 26/12/2013, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain the advantages of Multi-level Inverters. [04]
(b) State advantages of active power filters over passive harmonic filters. [03]
- Q - 2 (a) What is redundancy of switching states? Explain the operation for one leg of 3-level neutral point clamped inverter. [07]
(b) Explain in detail operation for one leg of 3-level flying capacitor inverter. Explain the balancing of clamping capacitors. [07]

OR

- Q - 2 (a) Explain in detail working of 3- Φ bridge inverter for 180° mode of conduction. Also, draw phase and line voltage waveforms for a star connected resistive load. [07]
(b) Explain in detail working of 12 -pulse converter. [07]
- Q - 3 (a) Explain briefly with neat diagram different types of active power filters. [07]
(b) Explain in detail operation of load commutated inverter supplying synchronous motor with line commutated converter as its current source. [07]

OR

- Q - 3 (a) What is the role of reference compensating current generation scheme for shunt active power filters? Explain instantaneous reactive power theory for generation of reference compensating currents. [07]
(b) Explain with neat voltage and current waveforms operation of 1- Φ , auto-sequentially commutated inverter supplying inductive load. [07]

SECTION - II

- Q - 4 (a) State advantages and disadvantages of Flexible AC Transmission system. [04]
(b) What is the need of combined shunt and series compensators? [03]
- Q - 5 (a) Explain in detail working of thyristor switched reactor and thyristor switched capacitor. [07]
(b) Write a short note on following: [07]
 - Thyristor controlled reactor (TCR)
 - Fixed capacitor, Thyristor controlled reactor (FC-TCR)

OR

- Q - 5 (a) Write a short note on following: [07]
- shunt compensation
 - comparison of different SVC
- (b) Explain in detail working of STATCOM. [07]
- Q - 6 (a) Explain the basic operating principle of Unified Power Flow Controller. Also, give [07]
- comparison of Unified Power Flow Controller with controlled series compensators.
- (b) Explain in detail working of thyristor switched series capacitor. [07]

OR

- Q - 6 (a) Explain basic operating principle and characteristics of Interline Power Flow [07]
- Controller.
- (b) Explain in detail working and control scheme of thyristor controlled series capacitor. [07]
